

UNIT -II

Data Mining Techniques – a Statistical Perspective on data mining – Similarity Measures – Decision Trees – Neural Networks – Genetic Algorithms.

INTRODUCTION[Data Mining Techniques]

There are many different methods used to perform data mining tasks. These techniques not only require specific types of data structures, but also imply certain types of algorithmic approaches. Parametric models describe the relationship between input and output through the use of algebraic equations where some parameters are not specified. These unspecified parameters are determined by providing input examples.

Non-parametric techniques are more appropriate for data mining applications. A non-parametric model is one that is data-driven. No explicit equations are used to data mining the model. This means that the modeling process adapts to the data at hand. Unlike parametric modeling, where a specific model is assumed ahead of time, the non-parametric techniques create a model based on the input. While the parametric methods require more knowledge about the data before the modeling process, the non-parametric technique requires a large amount of data as input to the modeling process itself.

Recent non-parametric methods have employed machine learning techniques to be able to learn dynamically as data are added to the input. These features make non-parametric techniques particularly suitable to database applications with large amounts of dynamically changing data. Non-parametric techniques include neural networks, decision trees, and genetic algorithms.

A Statistical Perspective on data mining:

There have been many statistical concepts that are the basis for data mining techniques.

Point Estimation

Point estimation refers to the process of estimating a population parameter by an estimate of the parameter,. This can be done to estimate mean, variance, standard deviation, or any other statistical parameter. Often the estimate of the parameter for a general population may be made by actually calculating the parameter value for a population sample.

An estimator technique may also be used to estimate (predict) the value of missing data. The bias of an estimator is the difference between the expected value of the estimator and the actual value:

$$\text{Bias} = E(\hat{\Theta}) - \Theta$$

An unbiased estimator is one whose bias is 0. While point estimators for small data sets may actually be unbiased, for larger database applications we would expect that most estimators are biased.

One measure of the effectiveness of an estimate is the mean squared error (MSE), which is defined as the expected value of the squared difference between the estimate and the actual value:

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \Theta)^2$$

The squared error is often examined for a specific prediction to measure accuracy rather than to look at the average difference.

For example, if the true value for an attribute was 10 and the prediction was 5, the squared error would be $(5 - 10)^2 = 25$.

The squaring is performed to ensure that the measure is always positive and to give a higher weighting to the estimates that are grossly inaccurate. The MSE is commonly used in evaluating the effectiveness of data mining prediction techniques. It is also important in machine learning.

At times, instead of predicting a simple point estimate for a parameter, one may determine a range of values within which the true parameter value should fall. This range is called a **confidence interval**.

The **root mean square (RMS)** may also be used to estimate error or as another statistic to describe a distribution. Calculating the mean does not indicate the magnitude of the values. The RMS can be used for this purpose.

Given a set of n values $X = \{x_1, \dots, x_n\}$, the RMS is defined by

$$\text{RMS} = \sqrt{\frac{\sum_{j=1}^n x_j^2}{n}}$$

NOTE: An alternative use is to estimate the magnitude of the error. The root mean square error (RMSE) is found by taking the square root of the MSE.

A popular estimating technique is the jackknife estimate. With this approach, the estimate of a parameter, $\hat{\theta}$, is obtained by omitting one value from the set of observed values. Suppose that there is a set of values $X = \{x_1, \dots, x_n\}$. An estimate for the mean would be

$$\hat{\mu}_{(i)} = \frac{\sum_{j=1}^{i-1} x_j + \sum_{j=i+1}^n x_j}{n - 1}$$

Here the subscript (i) indicates that this estimate is obtained by omitting the i^{th} value. Given a set of jackknife estimates, $\hat{\theta}_{(i)}$, these can in turn be used to obtain an overall estimate.

Another technique for point estimation is called the **maximum likelihood estimate** can be defined as a value proportional to the actual probability that with a specific distribution the given sample exists. So the sample gives us an estimate for a parameter from the distribution.

The higher the likelihood value, the more likely the underlying distribution will produce the results observed. Given a sample set of values $X = \{x_1, \dots, x_n\}$ from a known distribution function $f(x_i | \hat{\theta})$, the MLE can estimate parameters for the population from which the sample is drawn. The approach obtains parameter estimates that maximize the probability that the sample data occur for the specific model. It looks at the joint probability for observing the sample data by multiplying the individual probabilities.

The likelihood function, L , is thus defined as

$$L(\Theta | x_1, \dots, x_n) = \prod_{i=1}^n f(x_i | \Theta)$$

The value of $\hat{\Theta}$ that maximizes L is the estimate chosen. This can be found by taking the derivative (perhaps after finding the log of each side to simplify the formula) with respect to $\hat{\Theta}$.

Models Based on Summarization

There are many basic concepts that provide an abstraction and summarization of the data as a whole. The basic well-known statistical concepts such as mean, variance, standard deviation, median, and mode are simple models of the underlying population.

There are also many well-known techniques to display the structure of the data graphically. For example, a histogram shows the distribution of the data. A box plot is a more sophisticated technique that illustrates several different features of the population at once.

Figure 3.1 shows a sample box plot. The total range of the data values is divided into four equal parts called quartiles. The box in the center of the figure shows the range between the first, second, and third quartiles. The line in the box shows the median. The lines extending from either end of the box are the values that are a distance of 1.5 of the interquartile range from the first and third quartiles, respectively. Outliers are shown as points beyond these values.

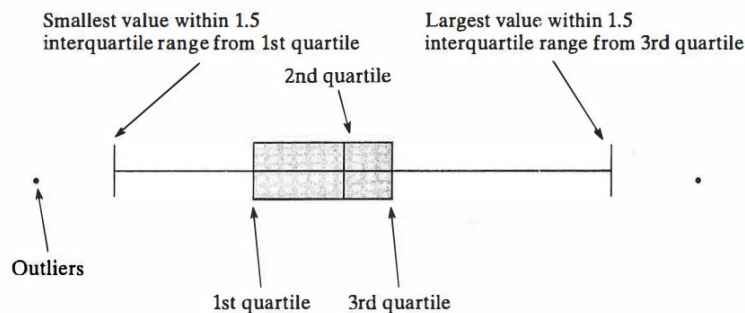


FIGURE 3.1: Box plot example.

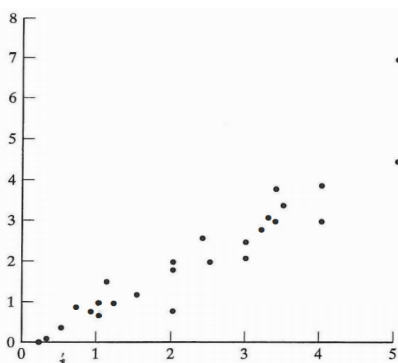


FIGURE 3.2: Scatter diagram example.

Another visual technique to display data is called a **scatter diagram**. This is a graph on a two-dimensional axis of points representing the relationships between x and y values. By plotting the actually observable (x, y) points as seen in a sample, a visual image of some derivable functional relationship between the x and y values in the total population may be seen.

Bayes Theorem

With statistical inference, information about a data distribution are inferred by examining data that follow that distribution.

Given a set of data $X = \{x_1, \dots, x_n\}$, a data mining problem is to uncover properties of the distribution from which the set comes. Bayes rule, defined in Definition 3.1, is a technique to estimate the likelihood of a property given the set of data as evidence or input. Suppose that either hypothesis h_1 or hypothesis h_2 must occur, but not both. Also suppose that X_i is an observable event.

DEFINITION 3.1. Bayes Rule or Bayes Theorem is

$$P(h_1 | x_i) = \frac{P(x_i | h_1)P(h_1)}{P(x_i | h_1)P(h_1) + P(x_i | h_2)P(h_2)}$$

Here $P(h_1 | x_i)$ is called the posterior probability, while $P(h_1)$ is the prior probability associated with hypothesis h_1 . $P(x_i)$ is the probability of the occurrence of data value X_i and $P(x_i | h_1)$ is the conditional probability that, given a hypothesis, the tuple satisfies it. Bayes rule allows us to assign probabilities of hypotheses given a data value, $P(h_j | X_i)$.

EXAMPLE

Suppose that a credit loan authorization problem can be associated with four

hypotheses: $H = \{h_1, h_2, h_3, h_4\}$ where h_1 = authorize purchase, h_2 = authorize after further identification, h_3 = do not authorize, and h_4 = do not authorize but contact police.

The training data for this example are shown in Table 3.1. From training data, we find that $P(h_1) = 60\%$, $P(h_2) = 20\%$, $P(h_3) = 10\%$, and $P(h_4) = 10\%$.

To make our predictions, a domain expert has determined that the attributes we should be looking at are income and credit category. Assume that income, I , has been categorized by ranges $[0, \$10,000)$, $[\$10,000, \$50,000)$, $[\$50,000, \$100,000)$, and $[\$100,000, INFINITE]$.

These ranges are encoded and are shown in Table as 1, 2, 3, and 4, respectively.

TABLE 3.1: Training Data for Example 3.3

ID	Income	Credit	Class	x_i
1	4	Excellent	h_1	x_4
2	3	Good	h_1	x_7
3	2	Excellent	h_1	x_2
4	3	Good	h_1	x_7
5	4	Good	h_1	x_8
6	2	Excellent	h_1	x_2
7	3	Bad	h_2	x_{11}
8	2	Bad	h_2	x_{10}
9	3	Bad	h_3	x_{11}
10	1	Bad	h_4	x_9

Suppose that credit is categorized as excellent, good, or bad. By combining these, we then have 12 values in the data space: $D = \{x_i, \dots, x_{12}\}$.

TABLE 3.2: x_i Assignments for Example 3.3

	1	2	3	4
Excellent	x_1	x_2	x_3	x_4
Good	x_5	x_6	x_7	x_8
Bad	x_9	x_{10}	x_{11}	x_{12}

The relationship between these x_i values and the two attributes is shown in Table 3.2. Using these values, the last column in Table 3.1 shows the x_i group into which that tuple falls.

Given these, we can then calculate $P(x_i | h_j)$ and $P(x_i)$. We illustrate how this is done with h_1 . There are six tuples from the training set that are in h_1 ; we use the distribution of these across the x_i to obtain: $P(x_7 | h_1) = 2/6$, $P(x_4 | h_1) = 1/6$, $P(x_2 | h_1) = 2/6$, $P(x_8 | h_1) = 1/6$, and $P(x_i | h_1) = 0$ for all other values of i . Suppose we wanted to predict the class for x_4 . We thus need to find $P(h_j | x_4)$ for each h_j . We would then classify x_4 to the class with the largest value for h_1 . We find $P(h_1 | x_4) = \frac{(P(x_4 | h_1))(P(h_1))}{P(x_4)} = \frac{(1/6)(0.6)}{0.1} = 1$. We would thus classify x_4 to the h_1 class.

This example illustrates some issues associated with sampling. First note that Table 3.1 has no entries for X_1 , X_3 , X_5 , X_6 , or X_{12} . This makes it impossible to use this training sample to determine how to make predictions for this combination of input data. If indeed these combinations never occur, this would not be a problem.

Hypothesis Testing

Hypothesis testing attempts to find a model that explains the observed data by first creating a hypothesis and then testing the hypothesis against the data. This is in contrast to most data mining approaches, which create the model from the actual data without guessing what it is first. The actual data itself drive the model creation. The hypothesis usually is verified by examining a data sample. If the hypothesis holds for the sample, it is assumed to hold for the population in general. Given a population, the initial (assumed) hypothesis to be tested, H_0 , is called the null hypothesis. Rejection of the null hypothesis causes another hypothesis, H_1 , called the alternative hypothesis, to be made.

One technique to perform hypothesis testing is based on the use of the chi-squared statistic. Actually, there is a set of procedures referred to as chi squared.

These procedures can be used to test the association between two observed variable values and to determine if a set of observed variable values is statistically significant (i.e., if it differs from the expected case). A hypothesis is first made, and then the observed values are compared based on this hypothesis.

Assuming that O represents the observed data and E is the expected values based on the hypothesis, the chi-squared statistic, χ^2 , is defined as:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

When comparing a set of observed variable values to determine statistical significance, the values are compared to those of the expected case. This may be the uniform distribution.

EXAMPLE

Suppose that there are five schools being compared based on students' results on a set of standardized achievement tests. The school district expects that the results will be the same for each school. They know that the total score for the schools is 375, so the expected result would be that each school has an average score of 75. The actual average scores from the schools are: 50, 93, 67, 78, and 87. The district administrators want to determine if this is statistically significant. Or in simpler terms, should they be worried about the distribution of scores. The chi-squared measure here is

$$\chi^2 = \frac{(50 - 75)^2}{75} + \frac{(93 - 75)^2}{75} + \frac{(67 - 75)^2}{75} + \frac{(78 - 75)^2}{75} + \frac{(87 - 75)^2}{75} = 15.55 \quad (3.24)$$

Examining a chi-squared significance table, it is found that this value is significant. With a degree of freedom of 4 and a significance level of 95%, the critical value is 9.488. Thus, the administrators observe that the variance between the schools' scores and the expected values cannot be associated with pure chance.

Regression and Correlation

Both bi-variate regression and correlation can be used to evaluate the strength of a relationship between two variables. Regression is generally used to predict future values based on past values by fitting a set of points to a curve. Correlation, however, is used to examine the degree to which the values for two variables behave similarly.

Linear regression assumes that a linear relationship exists between the input data and the output data. The common formula for a linear relationship is used in this model:

$$y = C_0 + C_1 X_1 + \dots + C_n + X_n$$

Here there are n input variables, which are called predictors or regressors; one output variable (the variable being predicted), which is called the response; and $n + 1$ constants, which are chosen during the modeling process to match the input examples (or sample). This is sometimes called multiple linear regression because there is more than one predictor.

Example this is an example of the use of linear regression.

It is known that a state has a fixed sales tax, but it is not known what the amount happens to be. The problem is to derive the equation for the amount of sales tax given an input purchase amount. We can state the desired linear equation to be $y = C_0 + C_1 + X_1$. So we really only need to have two samples of actual data to determine the values of C_0 and C_1 . Suppose that we know (10, 0.5) and (25, 1.25) are actual purchase amount and tax amount pairs. Using these data points, we easily determine that $C_0 = 0$ and $C_1 = 0.05$. Thus, the general formula is $y = 0.05 x_i$. This would be used to predict a value of y for any known x_i value.

SIMILARITY MEASURES

- The use of similarity measures is well known to anyone who has performed Internet searches using a search engine. In such a search, the set of all Web pages represents the whole database, and these are divided into two classes: those that answer your query and those that do not. Those that answer your query should be more like each other than those that do not answer your query. The similarity in this case is defined by the query you state, usually based on a keyword list. Thus, the retrieved pages are similar because they all contain (to some degree) the keyword list you have specified.
- The idea of similarity measures can be abstracted and applied to more general classification problems. The difficulty lies in how the similarity measures are defined and applied to the items in the database. Since most similarity measures assume numeric (and often discrete) values, they may be difficult to use for more general data types. A mapping from the attribute domain to the subset of the integers may be used.

DEFINITION . The similarity between two tuples t_i and t_j , $\text{sim}(t_i, t_j)$, in a database D is a mapping from $D \times D$ to the range $[0, 1]$. Thus, $\text{sim}(t_i, t_j) \in [0, 1]$. The objective is to define the similarity mapping such that documents that are more alike have a higher similarity value.

So how does one define such a similarity mapping? This, of course, is the difficult part. Often the concept of "alike" is itself not well defined. When the idea of similarity measure is used in classification where the classes are predefined, this problem is somewhat simpler

than when it is used for clustering where the classes are not known in advance. Again, think of the IR example. Each IR query provides the class definition in the form of the IR query itself. So the classification problem then becomes one of determining similarity not among all tuples in the database but between each tuple and the query. This makes the problem an $O(n)$ problem rather than an $O(n^2)$ problem. Here are some of the more common similarity measures used in traditional IR systems and more recently in Internet search engines:

- **Dice:** $\text{sim}(t_i, t_j) = \frac{2 \sum_{h=1}^k t_{ih} t_{jh}}{\sum_{h=1}^k t_{ih}^2 + \sum_{h=1}^k t_{jh}^2}$
- **Jaccard:** $\text{sim}(t_i, t_j) = \frac{\sum_{h=1}^k t_{ih} t_{jh}}{\sum_{h=1}^k t_{ih}^2 + \sum_{h=1}^k t_{jh}^2 - \sum_{h=1}^k t_{ih} t_{jh}}$
- **Cosine:** $\text{sim}(t_i, t_j) = \frac{\sum_{h=1}^k t_{ih} t_{jh}}{\sqrt{\sum_{h=1}^k t_{ih}^2 \sum_{h=1}^k t_{jh}^2}}$
- **Overlap:** $\text{sim}(t_i, t_j) = \frac{\sum_{h=1}^k t_{ih} t_{jh}}{\min \left(\sum_{h=1}^k t_{ih}^2, \sum_{h=1}^k t_{jh}^2 \right)}$

Distance or dissimilarity measures are often used instead of similarity measures. As implied, these measure how “unlike” items are. Traditional distance measures may be used in a two-dimensional space. These include

- **Euclidean:** $\text{dis}(t_i, t_j) = \sqrt{\sum_{h=1}^k (t_{ih} - t_{jh})^2}$
- **Manhattan:** $\text{dis}(t_i, t_j) = \sum_{h=1}^k |t_{ih} - t_{jh}|$

To compensate for the different scales between different attribute values, the attribute values may be normalized to be in the range [0, 1]. If nominal values rather than numeric values are used, some approach to determining the difference is needed. One method is to assign a difference of 0 if the values are identical and a difference of 1 if they are different.

DECISION TREES

A decision tree is a predictive modeling technique used in classification, clustering, and prediction tasks. Decision trees use a "divide and conquer" technique to split the problem search space into subsets.

It is based on the "Twenty Questions" game that children play, as illustrated by Example 3.6. Figure 3.4 graphically shows the steps in the game. This tree has as the root the first question asked. Each subsequent level in the tree consists of questions at that stage in the game. Nodes at the third level show questions asked

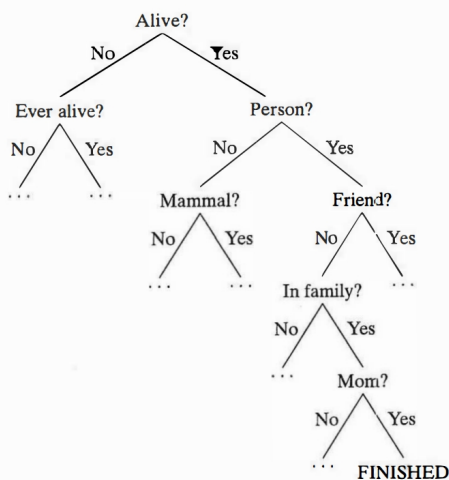


FIGURE 3.4: Decision tree for Example 3.6.

at the third level in the game. Leaf nodes represent a successful guess as to the object being predicted. This represents a correct prediction. Each question successively divides the search space much as a binary search does. As with a binary search, questions should be posed so that the remaining space is divided into two equal parts. Often young children tend to ask poor questions by being too specific, such as initially asking "Is it my Mother?" This is a poor approach because the search space is not divided into two equal parts.

EXAMPLE 3.6

Stephanie and Shannon are playing a game of "Twenty Questions." Shannon has in mind some object that Stephanie tries to guess with no more than 20 questions. Stephanie's first question is "Is this object alive?" Based on Shannon's answer, Stephanie then asks a second question. Her second question is based on the answer that Shannon provides to the first question. Suppose that Shannon says "yes" as her first answer. Stephanie's second question is "Is this a person?" When Shannon responds "yes," Stephanie asks "Is it a friend?" When Shannon says "no," Stephanie then asks "Is it someone in my family?" When Shannon responds "yes," Stephanie then begins asking the names of family members and can immediately narrow down the search space to identify the target individual. This game is illustrated in Figure 3.4.

A **decision tree (DT)** is a tree where the root and each internal node is labeled with a question. The arcs emanating from each node represent each possible answer to the associated question. Each leaf node represents a prediction of a solution to the problem under consideration.

NEURAL NETWORKS

- The first proposal to use an artificial neuron appeared in 1943, but computer usage of neural networks did not actually begin until the 1980s.
- Neural networks (NN), often referred to as artificial neural networks (ANN) to distinguish them from biological neural networks, are modeled after the workings of the human brain.
- The NN is actually an information processing system that consists of a graph representing the processing system as well as various algorithms that access that graph. As with the human brain, the NN consists of many connected processing elements.
- The NN, then, is structured as a directed graph with many nodes (processing elements) and arcs (interconnections) between them. The nodes in the graph are like individual neurons, while the arcs are their interconnections.
- Each of these processing elements functions independently from the others and uses only local data (input and output to the node) to direct its processing. This feature facilitates the use of NNs in a distributed and/or parallel environment.

The NN approach, like decision trees, requires that a graphical structure be built to represent the model and then that the structure be applied to the data. The NN can be viewed as a directed graph with source (input), sink (output), and internal (hidden) nodes. The input nodes exist in a input layer, while the output nodes exist in an output layer. The hidden nodes exist over one or more hidden layers.

To perform Data mining task, a tuple is input through the input nodes and the output node determines what the prediction is. Unlike decision trees, which have only one input node (the root of the tree), the NN has one input node for each attribute value to be examined to solve the data mining function. Unlike decision trees, after a tuple is processed, the NN may be changed to improve future performance. Although the structure of the graph does not change, the labeling of the edges may change.

As with decision trees, we define a neural network in two parts:

- one for the data structure
- one for the general approach used, including the data structure and the algorithms needed to use it.

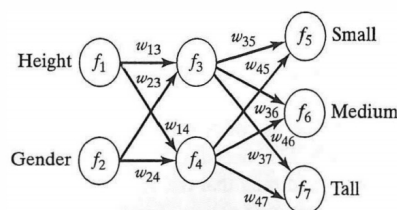


FIGURE 3.6: Neural network for Example 3.7.

DEFINITION A neural network (NN) model is a computational model consisting of three parts:

1. Neural network graph that defines the data structure of the neural network.
2. Learning algorithm that indicates how learning takes place.
3. Recall techniques that determine how information is obtained from the network. We discuss propagation in this text.

- NNs have been used in pattern recognition, speech recognition and synthesis, medical applications (diagnosis, drug design), fault detection, problem diagnosis, robot control, and computer vision.
- In business, NNs have been used to "advise" booking of airline seats to increase profitability.
- As a matter of fact, NNs can be used to compute any function. Although NNs can solve problems that seem more elusive to other AI techniques, they have a long training time (time during which the learning takes place) and thus are not appropriate for real-time applications. NNs may contain many processing elements and thus can be used in massively parallel systems.
- Artificial NNs can be classified based on the type of connectivity and learning. The basic type of connectivity discussed in this text is feed forward, where connections are only to layers later in the structure. Alternatively, a NN may be feedback where some links are back to earlier layers. Learning can be either supervised or unsupervised.

Activation Functions

The output of each node i in the NN is based on the definition of a function f_i , activation function, associated with it. An activation function is sometimes called a processing element function or a squashing function. The function is applied to the set of inputs coming in on the input arcs. Figure 3.7 illustrates the process. There have been many proposals for activation functions, including threshold, sigmoid, symmetric sigmoid, and Gaussian.

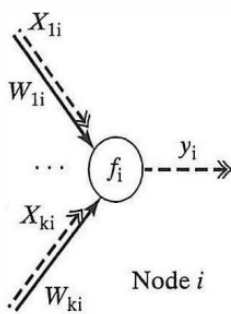


FIGURE 3.7: Neural network node.

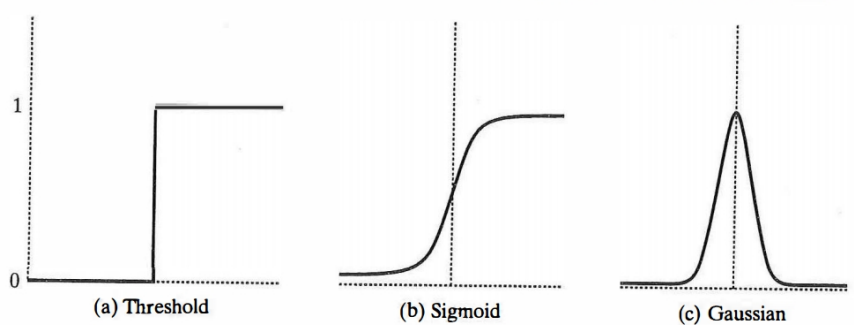


FIGURE 3.8: Sample activation functions.

- An activation function may also be called a firing rule, relating it back to the workings of the human brain.
- When the input to a neuron is large enough, it fires, sending an electrical signal out on its axon (output link).
- Likewise, in an artificial NN the output may be generated only if the input is above a certain level; thus, the idea of a firing rule.
- When dealing only with binary output, the output is either 0 or 1, depending on whether the neuron should fire. Some activation functions use -1 and 1 instead, while still others output a range of values. Based on these ideas, the input and output values are often considered to be either 0 or 1.

Genetic Algorithm

In data mining, genetic algorithms may be used for clustering, prediction, and even association rules. You can think of these techniques as finding the "fittest" models from a set of models to represent the data. In this approach a starting model is assumed and through many iterations, models are combined to create new models. The best of these, as defined by a fitness function, are then input into the next iteration. Algorithms differ in how the model is represented, how different individuals in the model are combined, and how the fitness function is used.

- In genetic algorithms, reproduction is defined by precise algorithms that indicate how to combine the given set of individuals to produce new ones. These are called crossover algorithms.
- Given two individuals (parents from the population, the crossover technique generates new individuals (offspring or children) by switching sub-sequences of the strings.
- Since genetic algorithms attempt to model nature, only the strong survive. When new individuals are created, a choice must be made about which individuals will survive.
- This may be the new individuals, the old ones, or more likely a combination of the two. The third major component of genetic algorithms, then, is the part that determines the best (or fittest) individuals to survive.
- One of the most important components of a genetic algorithm is determining how to select individuals. A fitness function, f , is used to determine the best individuals in a population. This is then used in the selection process to choose parents: Given an objective by which the population can be measured, the fitness function indicates how well the goodness objective is being met by an individual.

DEFINITION A genetic algorithm (GA) is a computational model consisting of five parts:

1. Starting set of individuals, P .
2. Crossover technique.
3. Mutation algorithm.
4. Fitness function
5. Algorithm that applies the crossover and mutation techniques to P interactively using the fitness function to determine the best individuals in P to keep. The algorithm replaces a predefined number of individuals from the population with each iteration and terminates when some threshold is met.

Genetic algorithms have been used to solve most data mining problems, including classification, clustering, and generating association rules. Typical applications of genetic algorithms include scheduling, robotics, economics, biology, and pattern recognition.

The major advantage to the use of genetic algorithms is that they are easily parallelized. There are, however, many disadvantages to their use:

- Genetic algorithms are difficult to understand and to explain to end users.
 - The abstraction of the problem and method to represent individuals is quite difficult.
 - Determining the best fitness function is difficult.
 - Determining how to do crossover and mutation is difficult.
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What is a Genetic Algorithm?

Genetic algorithms (GA) are adaptive search algorithms- adaptive in terms of the number of parameters you provide or the types of parameters you provide. The algorithms classify the best optimal solution among the several solutions, and its design is based on the natural genetic solution.

Genetic algorithm emulates the principles of natural evolution, i.e. survival of the fittest. Natural evolution propagates the genetic material in the fittest individuals from one generation to the next.

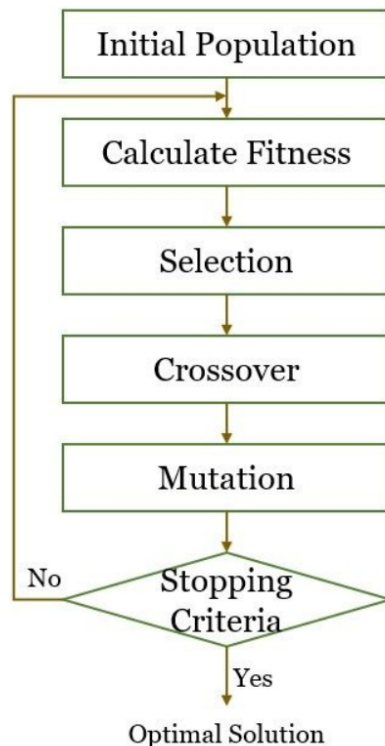
The genetic algorithm applies the same technique in data mining – it iteratively performs the selection, crossover, mutation, and encoding process to evolve the successive generation of models.

The components of genetic algorithms consist of:

- Population incorporating individuals.
- Encoding or decoding mechanism of individuals.
- The objective function and an associated fitness evaluation criterion.
- Selection procedure.
- Genetic operators like recombination or crossover, mutation.
- Probabilities to perform genetic operations.
- Replacement technique.
- Termination combination.

At every iteration, the algorithm delivers a model that inherits its traits from the previous model and competes with the other models until the most predictive model survive.

If we count the phases of the genetic algorithm, then we would distribute the entire algorithm into six phases, as you can see in the image below:



Initial population

Being the first phase of the algorithm, it includes a set of individuals where each individual is a solution to the concerned problem. We characterize each individual by the set of parameters that we refer to as **genes**.

Calculate Fitness

A fitness function is implemented to compute the fitness of each individual in the population. The function provides a fitness score to each individual in the population. The fitness score is the probability of the individual selection in the reproduction process.

Selection

The selection process selects the individuals with the highest fitness score and is allowed to pass on their genes to the next generation.

Crossover

It is a core phase of the genetic algorithm. Now the algorithm chooses a crossover point within the parents' genes chosen for mating.

The algorithm keeps generating the offspring until the groups of parents exchange their genes until they reach the crossover point. Now, these newly created offspring are added to the population.

Mutation

The mutation phase inserts random genes into the generated offspring to maintain the population's diversity. It is done by flipping random genes in new offspring.

Termination

The iteration of the algorithm stops when it produces offspring that is not different from the previous generation. It is said to have produced a set of solutions for the problem at this stage.

Working of Genetic algorithm with Example

To overview the evolution of models until the best optimal solution occurs, we will discuss a simple example.

Suppose a company wants to increase its profit, so it came up with the idea of sending promotional mail along with the coupons. This promotional mail must increase the sales and profit of the company; however, it can also produce a negative result of revenue loss.

So here, the genetic algorithm has to come up with the solution that what optimum number of coupons must be added to each promotional mail to maximize the company's profit.

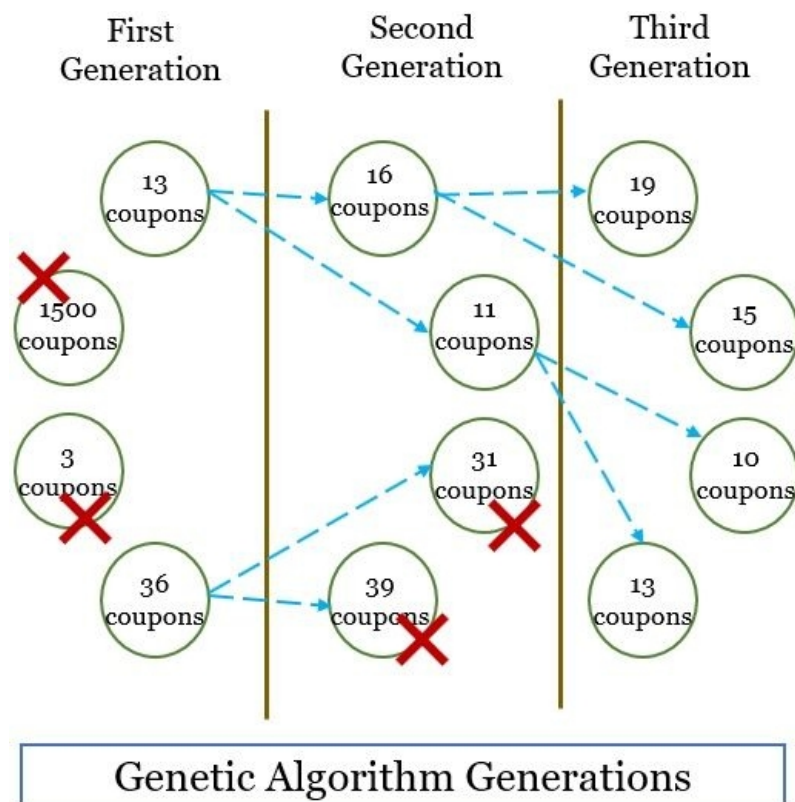
The initial view would be to add as many coupons as possible to maximize the profit. This will let the customer use all the coupons and maximize the company's profit.

Well, some factors may complicate the idea of adding coupons to the mail to maximize the profit, such as:

- Adding more coupons in the mail would increase the postal cost, ultimately reducing the profit.
- Sending fewer coupons will also reduce the opportunity to gain profit and lead to a potential loss in revenue.
- Adding too many coupons may reduce the customer's interest in using any coupon at all.

In such a case, we need an optimum number of coupons that must be sent in each mail to maximize the profit.

The image below represents the evolution obtained by implementing the genetic algorithm for this problem. Over the three generations, we have obtained the optimum number of coupons that must be sent in each mail.



You can notice that in every generation, the fittest option survives. The generation keeps on evolving until the generation represents the optimal number of coupons per mail.

Although it is a very basic example, we have not discussed how these numbers are generated at each iteration. Nor we have discussed how we have eliminated the abnormal numbers. We have complex calculations to perform these kinds of functions. Hopefully, the example gives you a basic idea of how the genetic algorithm works.

Genetic Algorithms in Data Mining

So far, we have studied that the genetic algorithm is a classification method that is adaptive, robust and used globally in situations where the area of classification is large. The algorithms optimize a fitness function based on the criteria preferred by data mining so as to obtain an optimal solution, for example:

- Knowledge discovery system
- MASSON system

However, the data mining application based on the genetic algorithm is not as rich as the application based on fuzzy sets. In the section ahead, we have categorized some systems based on the genetic algorithm used in data mining.

Regression

Data mining identifies human interpretable patterns; it includes a prediction that determines a future value from the available variable or attributes in the database. The basic assumption of the linear multi regression model is that there is no interaction among the attributes.

GA handles the interaction among the attributes in a far better way. The non-linear multi regression model uses GA to get single out from the training data set.

Association Rules

Multiple objective GA deals with the problems with multiple objective functions and constraints, to determine the optimal set of solutions. None of the solutions from this set must exist in the search space that can dominate any member of this set.

Such algorithms are used for rule mining with a large search space with many attributes and records. To obtain the optimal solutions, multi-objective GA performs the global search with multiple objectives. Such as a combination of factors like predictive accuracy, comprehensibility and interestingness.

Advantages and Disadvantages

Advantages

- Easy to understand as it is based on the concept of natural evolution.
- Classifies an optimal solution from a set of solutions.
- GA uses the pay off information instead of the derivative to yield an optimal solution.
- GA backs multi-objective optimization.
- GA is an adaptive search algorithm.
- GA also operates in a noisy environment.

Disadvantages

- An improper implementation may lead to a solution that is not optimal.
- Implementing fitness function iteratively may lead to computational challenges.
- GA is time-consuming as it deals with a lot of computation.

Applications of GA

GA is used in implementing many applications let s discuss a few of them.

- Economics: In the field of economics GA is used to implement certain models that conduct competitive analysis, decision making, and effective scheduling.
- Aircraft Design: GA is used to provide the parameters that must be modified and upgraded in order to get a better design.
- DNA Analysis: GA is used to establish DNA structure using spectrometric information.
- Transport: GA is used to develop a transport plan that is time and cost-efficient.
- Data mining: GA classify a large set of data to determine the optimal solution to the concerned problem.